

Anonymity Contract Objects for Anonymous DHTs: A Practical Interface for Proof-Carrying Claims (Series Synthesis)

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Abstract

The papers in this archive share a design goal: *brief, referee-grade* anonymity arguments that can be certified and updated without shipping a monolithic research archive. To keep redundancy low while staying practical, we treat each mechanism paper as exporting a small *contract* in a standard format: a witness family, a metric, a budget, and an accountant describing repeats/resets. This note proposes a concrete “contract object” interface for anonymous overlays and anonymous DHT lookups, and shows how it connects the archive’s main endpoints: (i) odds-inflation / pointwise-maximal-leakage budgets, (ii) total-variation (TV) equalization contracts (including congestion witnesses), and (iii) notarized and auditable evaluation receipts under retries and many-testing; the centralized log-audit certificate note is Synthesis 25.[20] (see the receipt-accounting and conditional-certificate notes, Synthesis 19–21).[5, 18, 19] We give simple composition rules that justify “multi-tier” budgeting and clarify how to cross-pollinate insights across papers without re-proving the same facts.

Scope. This synthesis note defines the shared *interface* between mechanism proofs, endpoint metrics, and certificate/evaluation machinery. It is intentionally lightweight: we introduce only the contract-object vocabulary needed to keep the rest of the papers short and non-redundant.

Audience note (scope limitation). This series is written for deployers, reviewers, and auditors: it specifies how *providers* can make anonymity claims machine-checkable and how third parties can verify those claims from public receipts. End-user verification without access to receipts (e.g., adversarially testing a live provider from the outside) is important but out of scope here; we treat it as an additional threat model and interface problem. A minimal user-facing discoverability/validation interface is sketched in Synthesis 13 as an optional adjunct.[17]

Imports. Concrete mechanisms (scheduling/stop-time padding, TV toolboxes, congestion witnesses) and endpoint bridges (odds inflation / MaxL) are imported from the corresponding series papers and treated as black boxes. Trace-interface vocabulary (tiers, exposure normal forms, committee-contact trace schema) is owned by the trace-interfaces note (Synthesis 3), while notation conventions and the minimal probabilistic model are owned by the notation sheet (Synthesis 8).[15, 16]

Exports. We export (i) a contract-object schema (Def. 1), (ii) two micro-lemmas that appear in many receipts (tier additivity and union-style tail composition), and (iii) a compact conversion lemma that links odds-inflation style caps to multiplicative and TV-style bounds. Horizon- n accounting across repeats, retries, and state is owned by the receipt-accounting rulebook (Synthesis 19).[5]

Artifact policy. This archive is source-only; supporting artifacts and tooling are available on request.[14]

1 Where this fits in the series

The archive has three recurring themes:

1. **Compiler papers** (how to hide metadata): scheduling/stop-time mechanisms, delegation bounds, and TV-toolboxes.
2. **Endpoint metrics** (what the guarantee should *mean*): odds inflation / pointwise maximal leakage (Anonymity A) and TV-style contracts (Congestion-EQ / Synthesis 5).
3. **Certification and change control** (how to ship and update a claim): ABOM/OINL (Anonymity C), proof-carrying budgets (Boss Fight series), and notarized/auditable evaluation (evaluation series).

Redundancy is minimized if mechanism papers *export* a contract in a shared vocabulary, and endpoint/certification papers provide generic conversion and composition theorems.

2 A contract-object vocabulary

Definition 1 (Anonymity contract object). *An anonymity contract object for a mechanism $\mathcal{M}$ consists of:*

1. **Secret scope** S and **context** Π (a prior family on S , plus public eligibility interfaces).
2. **Witness family** $\mathcal{W}$: a declared set of observables (e.g. stop time, contact trace, congestion witness, retry transcript), together with an exposure normal-form id / schema hash naming the projection being budgeted.[15]
3. **Metric** $\mathbf{M}$ and **budget** B : a per-witness bound, typically one of:
 - **Odds-inflation / PML**: $\sup_{w \in \mathcal{W}} \max_{s \in S} \log_2 \frac{P_s(w)}{P(w)} \leq \varepsilon$ (Anonymity A).
 - **Hockey-stick**: $P_s(A) \leq 2^\varepsilon P(A) + \delta$ for all events A (Anonymity A / Anonymity B).
 - **TV equalization**: $\text{TV}(P_s, P) \leq \tau$ (compiler/TV toolbox papers; Congestion-EQ).
4. **Accountant** $\mathbf{A}$: an explicit rule for repeats/resets (epochs, retries, cache updates), producing a cumulative budget B_{tot} .
5. **Knob interface** $\mathbf{K}$: named parameters the deployer may choose (timeouts, padding distributions, committee size, calendar period), plus admissible ranges.
6. **Certificate and receipt schemas** (optional): machine-checkable claims, proofs, and monitoring artifacts (Anonymity C; Boss Fight; evaluation series). When a receipt line item points at a replay workflow, it should include not only a human plan name but also a digest-bound **plan_spec_id** committing to the canonical replay-plan specification (Synthesis 12; worked example Synthesis 17).

Remark 1 (Why this particular split). *The witness family is what prevents “metric laundering” (changing what is observed) and keeps proofs honest. The accountant is what prevents “repeat laundering” (replaying small leaks until they become large). Separating $\mathbf{K}$ from the proof lets a paper stay short: the paper proves “if knobs satisfy these inequalities, then the contract holds”; a deployment certificate instantiates the inequalities for its chosen knobs.*

3 Two basic composition rules that show up everywhere

Proposition 1 (Additive composition for odds inflation under conditional independence). *Let S be the secret and let W_1, W_2 be two witnesses such that conditional on $S = s$, the outputs are independent: $P_s(w_1, w_2) = P_s^{(1)}(w_1) P_s^{(2)}(w_2)$. Let $P^{(j)}$ be the public marginals induced by the same prior family. If each witness satisfies a pointwise odds-inflation cap*

$$\max_s \log_2 \frac{P_s^{(j)}(w_j)}{P^{(j)}(w_j)} \leq \varepsilon_j \quad \text{for all } w_j,$$

then the combined witness satisfies

$$\max_s \log_2 \frac{P_s(w_1, w_2)}{P(w_1, w_2)} \leq \varepsilon_1 + \varepsilon_2 \quad \text{for all } (w_1, w_2).$$

Equivalently, pointwise maximal leakage adds under conditionally independent tiers.

Proof. Under conditional independence, both the conditional laws and the public marginals factor. Therefore the likelihood ratio factors, and the log-likelihood ratio is a sum. $\square$

Proposition 2 (Union-style composition for tail budgets). *Suppose for each tier j we have a hockey-stick budget $P_s(A) \leq 2^{\varepsilon_j} P(A) + \delta_j$ for all events A over witness W_j . Then for the joint witness $W = (W_1, \dots, W_m)$,*

$$P_s(A) \leq 2^{\sum_j \varepsilon_j} P(A) + \sum_j \delta_j \quad \text{for all events } A \text{ over } W,$$

whenever the witness channels are conditionally independent given S .

4 A small conversion lemma that reduces redundancy

Several papers in the archive switch between “odds-inflation / likelihood-ratio” caps and TV-style contracts. The following one-line implication is a high-leverage cross-link.

Proposition 3 (Odds-inflation cap implies multiplicative and TV bounds). *Fix a prior-induced public marginal P and a secret-conditioned witness law P_s . If for all witnesses w we have the pointwise cap*

$$\log_2 \frac{P_s(w)}{P(w)} \leq \varepsilon,$$

then for every event A over the witness,

$$P_s(A) \leq 2^\varepsilon P(A).$$

Moreover, the total-variation distance satisfies the one-sided bound

$$\text{TV}(P_s, P) \leq 1 - 2^{-\varepsilon}.$$

(For small ε , this is $\text{TV}(P_s, P) \lesssim (\ln 2) \varepsilon$.)

Proof. The event bound is immediate by summing $P_s(w) \leq 2^\varepsilon P(w)$ over $w \in A$. For TV, write $\text{TV}(P_s, P) = \frac{1}{2} \int |f - 1| dP$ where $f = dP_s/dP$ satisfies $0 \leq f \leq 2^\varepsilon$ and $\int f dP = 1$. The maximum of $\int |f - 1| dP$ under these constraints is attained by a two-point extremal $f \in \{0, 2^\varepsilon\}$, yielding $\text{TV}(P_s, P) \leq 1 - 2^{-\varepsilon}$. $\square$

Remark 2 (Editorial rule). *If a paper proves a pointwise odds-inflation cap, it should not re-prove event/TV consequences. Instead, it should cite Proposition 3 (or the endpoint paper) and state the converted parameter values.*

Remark 3 (What this buys you in practice). *These two propositions justify a common, practical editorial move across the archive: if a paper introduces a new witness tier (congestion, retries, state evolution), it should not re-define “anonymity from scratch.” It should specify (i) the tier’s witness family and (ii) how its budget adds into the global accountant.*

5 How the main certificate objects line up

This paper is about the *contract-object interface*, not a full object catalog. To keep redundancy low, we maintain the authoritative, archive-wide object map in the top-level **SERIES_INDEX** (PDF) and in the math-backbone crosswalk (Synthesis 24).[4]

In brief, most papers in this archive export exactly one of three receipt-facing shapes:

- a **projection-level budget** (identified by `exposure_nf_id`) with an accountant over the declared threat window (`tw_id`);
- a **certificate object** that commits knobs and evidence by digest (ABOM/OINL; Certified series; evaluation receipts);
- a **guard object** that pins assumptions (state declarations, separation manifests) so drift is diffable.

Papers should cite the owning notes for these interfaces (Synthesis 3,4,8,12,18–21,22–23) rather than re-listing object fields here.

6 Cross-pollination opportunities, stated as reusable lemmas

In the unified archive, the following cross-links are the highest leverage (least redundancy, most reuse):

1. **Spectral** $\leftrightarrow$ **calendars/dither**. Spectral delegation bounds imply limits for periodic calendars; declared dither in PSC-Q can be analyzed as a spectral filter. A single lemma can be reused: “dither kernel contracts TV and attenuates low-frequency witnesses.”
2. **State evolution** $\leftrightarrow$ **certification**. State-dependent anonymity turns public scores/caches into witnesses; Closed-View Auditing turns witnesses into monitored receipts. The shared lemma is “small per-epoch bias amplifies quadratically under repetition,” motivating explicit accountants.
3. **Release-property ledgers** $\leftrightarrow$ **notarization**. Both are anti-equivocation devices: the ledger receipt and the notarized evaluation log differ mainly in what they commit to (software release vs. evaluation run). They can share an “append-only transparency log” interface and a standard auditor script.
4. **Audit objects are closed-view channels**. Any published receipt (PTL checkpoints, CPPC cards, release-ledger entries) is a *view* $Y = g(X)$ of some richer internal state/trace X . Treat g as a channel and budget it like any other witness: if g contracts TV by coefficient η (Dobrushin / SDPI), then a TV budget δ on X implies a TV budget at most $\eta\delta$ on the published view.

This is the common language behind (i) closed-view auditing in the state series, (ii) receipt-redaction bounds (Synthesis 1), and (iii) “publish what you account” design in the evaluation and release-ledger papers.

7 Bibliography (internal pointers)

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